Intro2Astro: Exoplanet Atmospheres Problem Set

July 29, 2025

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Section I: Life & Habitability

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Life and Habitability

a)

$$T_{eq} = \left(\frac{(1-A)L_{\odot}}{16\pi\sigma a^2} \right)^{1/4}$$

$A = 0.3$

$L = 3.827 \times 10^{26}\ W$

$\sigma = 5.67 \times 10^{-8}\ W m^{-2} K^{-4}$

a = habitable zone?

$$a = \left(\frac{(1-A)L_{\odot}}{16\pi\sigma T_{eq}^4} \right)^{1/2}$$

Inner Edge ($T = 373k$):

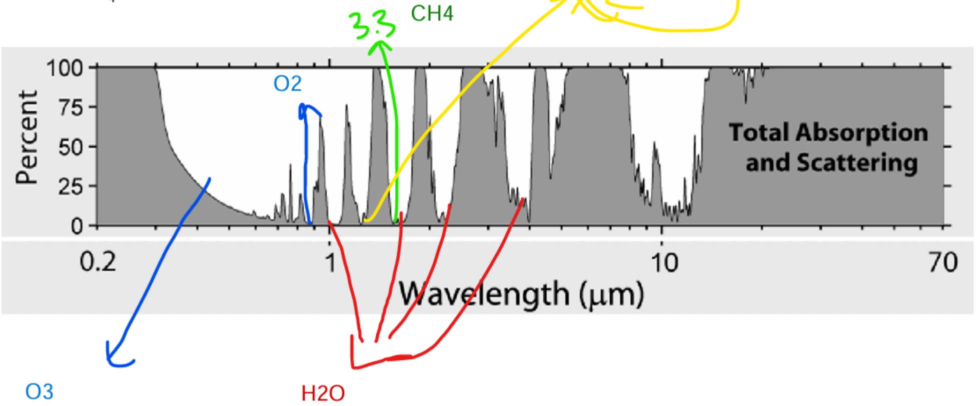
$a = 0,465$ AU

Outer Edge ($T = 273k$): $a = 0.87$ AU

b) This toy model underestimates the habitable zone. Earth orbits at 1 AU and is habitable, but the model gives an outer limit of 0.87 AU. This is because it ignores atmospheric effects like the greenhouse effect, which help retain heat.

Section II: Interpreting Atmospheric Absorption Spectra

reaching our ground-based detector. The figure below shows the total absorption of stellar radiation at different wavelengths for Earth's atmosphere as seen on the planet's surface. Label the major absorption features on the figure, including H_2O, CO_2, O_2 and O_3, and CH_4.



Section III: Characterizing Atmospheric Loss

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Life and Habitability

a) $$T_{eq} = \left(\frac{(1-A)L_\odot}{16\pi\sigma a^2}\right)^{1/4}$$

$A = 0.3$

$L = 3.827 \times 10^{26}$ W

$\sigma = 5.67 \times 10^{-8}$ W m^{-2} K^{-4}

a = habitable zone ?

$$a = \left(\frac{(1-A)L_\odot}{16\pi\sigma T_{eq}^4}\right)^{1/2}$$

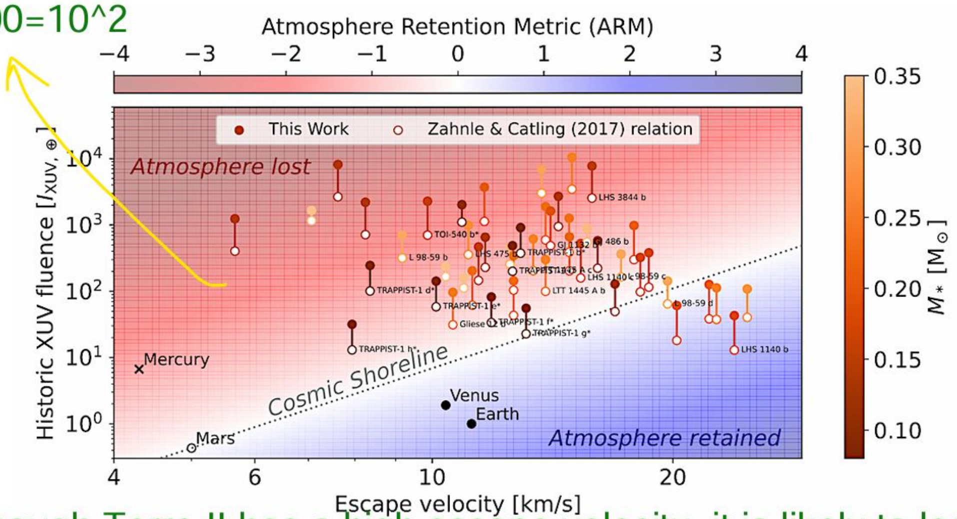
Inner Edge ($T = 373$ K):
$a = 0.465$ AU

Outer Edge ($T = 273$ K): $a = 0.87$ AU

b) This toy model underestimates the habitable zone. Earth orbits at 1 AU and is habitable, but the model gives an outer limit of 0.87 AU. This is because it ignores atmospheric effects like the greenhouse effect, which help retain heat.

for you to make your prediction. *(Hint: The XUV influence on Terra II is stated previously and read the escape velocity in units of km/s).*

x=15.81
y=100=10^2



Although Terra II has a high escape velocity, it is likely to lose its atmosphere due to receiving extremely high XUV radiation.